

EDIOMO UBONG ESU

Data Scientist (Early Career) | Energy Regulatory Analytics | AI/ML & Computer Vision | Statistical Modelling

Abuja, Nigeria • Open to Remote / Relocation •

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PROFESSIONAL SUMMARY

Early-career Data Scientist with a strong foundation in mathematics, statistics, and computing, applying machine learning, computer vision, and statistical modelling to real-world problems in Nigeria's oil and gas sector. Currently serving a one-year NYSC placement at Nigeria's Midstream and Downstream Petroleum Regulator (NMDPRA), where rotations across the Finance & Accounts Directorate have given first-hand exposure to how budget, account, and transaction data supports regulatory reporting and financial oversight in the energy value chain. Works in Python and SQL across the full modelling cycle, with self-directed projects in production forecasting, financial-integrity monitoring, and computer-vision-based pipeline and asset integrity inspection. Eager to keep growing professionally through mentorship, structured training, and hands-on work alongside experienced teams; open to roles, scholarships, fellowships, and internships in energy data analytics, energy-sector digitalization, and applied research, locally and internationally.

CORE COMPETENCIES

- Energy / Oil & Gas Data Analytics
- Machine Learning & Predictive Modelling
- Anomaly & Fraud Detection
- Model Evaluation & Selection
- Python Programming & SQL
- Experiment Design (A/B Testing)
- Dashboarding & Automated Reporting
- Regulatory Compliance & Reporting
- Production & Demand Forecasting
- Feature Engineering
- Statistical Modelling & Hypothesis Testing
- Data Preprocessing & Governance
- Digital Twin & Asset-Monitoring Concepts
- Insight Communication for Executive/Non-Technical Audiences

TECHNICAL SKILLS

Programming: Python, SQL, Java, C, MySQL

Machine Learning: Supervised & unsupervised learning, feature engineering, model building, evaluation & selection, optimization, end-to-end ML workflows

Computer Vision: Image preprocessing & augmentation, CNNs, transfer learning, image classification & object detection, visual defect/anomaly detection, Grad-CAM explainability, OpenCV

Statistical Modelling: Predictive analytics, probability, hypothesis testing, regression, time-series forecasting, A/B testing

Mathematical Foundations: Linear algebra, calculus & differential equations, optimization, numerical analysis, operations research

Energy Sector / Domain Knowledge: Oil & gas value-chain data, midstream/downstream regulatory reporting, budget & account compliance monitoring, production & finance data structures

Data Engineering (Basic) & Prep: Data preprocessing & cleaning, EDA, data mining & extraction, large-scale dataset management, data governance

Tools & Ecosystem: Python data-science stack, Jupyter/IDEs, Git, relational databases, Power BI, Tableau, Excel, dashboards

PROFESSIONAL EXPERIENCE

Data Science Intern – NYSC Corps Member, Energy Regulatory Analytics

Jan 2026 – Dec. 2026

Nigerian Midstream & Downstream Petroleum Regulatory Authority (NMDPRA) | Abuja, Nigeria

Finance & Accounts Directorate

- Serving a one-year national service placement, rotating across the Finance & Accounts Directorate's units, that is; Regional Accounts, Budget, Account System, Final Account, and Treasury, to learn how financial and operational data flows across a national energy regulator.

- Assisted with consolidating and reconciling financial data submitted by regional/zonal offices, and supported budget tracking and variance checks against approved allocations under the guidance of unit officers.
- Helped ensure transaction records were accurately captured and coded in the core accounting system, supported year-end account preparation with basic consistency checks, and assisted with cash-flow and disbursement monitoring in Treasury.
- Used Python to explore sample budget and transaction data for patterns and anomalies, and prepared summary dashboards and reports that presented the findings to supervisors in an accessible form.
- Gained first-hand insight into regulatory reporting and financial-oversight processes in the oil & gas value chain, which directly shaped the self-directed projects below.

Data Analyst & ICT Operations Manager

Sep 2025 – Apr 2026

Notre Dame Shalom School, Abuja, Nigeria

- Analyzed student academic performance data i.e. test scores, attendance, and grades to produce termly reports for school management, highlighting trends and flagging areas needing intervention.
- Digitized and organized student and administrative records into structured spreadsheets/databases, improving data accuracy, consistency, and retrieval across the school.
- Managed the school's ICT infrastructure computers, networks, and software systems troubleshooting issues and keeping day-to-day operations running smoothly.
- Supported staff with basic use of digital reporting tools, helping embed a more data-informed approach to academic and administrative decisions.

Mathematics & ICT Educator

Sep 2024 – Aug 2025

Wells International School, Akwa Ibom, Nigeria

- Taught mathematics and physics to secondary school students, covering algebra, calculus, trigonometry, and statistics, and developed lesson materials to strengthen quantitative reasoning.
- Served as Head of ICT/Science Clubs, introducing students to coding fundamentals and robotics projects to build early computational-thinking skills.
- Managed the school's ICT lab and equipment, ensuring functionality for both classroom instruction and extracurricular club activities.
- Mentored students preparing for internal mathematics and science assessments, reinforcing core concepts through structured practice sessions.

SELECTED PROJECTS - ENERGY & APPLIED DATA SCIENCE

Applied and self-directed projects reflecting direct engagement with data-science problems in the oil & gas and energy sector. Each is presented with its objective, method, and outcome; results are illustrative of the analytical approach and are being finalized with quantified metrics.

1. Anomaly Detection & Predictive Maintenance Model for Energy Infrastructure

Objective: To flag early signs of equipment and process anomalies in oil & gas infrastructure (pipelines, storage tanks, processing units), supporting a shift from reactive to predictive maintenance.

- Curated and preprocessed operational sensor time-series (pressure, flow, throughput proxies) and applied statistical and unsupervised anomaly-detection techniques to flag abnormal operating patterns.
- Delivered a working prototype showing how operators and regulators can move toward proactive, condition-based maintenance monitoring and reduce the risk of unplanned downtime.

Dataset: Volve Field Open Dataset (Equinor) real well-sensor and drilling time-series data (pressure, flow, choke settings) from the Volve oil field, North Sea, 2008–2016.

Tools & Methods: Python, pandas, scikit-learn, time-series analysis, statistical outlier detection

2. AI-Based Pipeline Defect Detection (Computer Vision)

Objective: To apply deep-learning computer vision to the detection and classification of surface defects, that is, corrosion, cracking, weld flaws, and leak indicators, on pipeline and storage-asset imagery, reducing reliance on slow, subjective manual visual inspection.

- Built an image-processing pipeline with resizing, normalization, and augmentation (rotation, flipping, brightness and contrast shifts) so the model generalizes across the varied lighting and weather conditions of field inspection.
- Fine-tuned pre-trained CNN backbones (ResNet / EfficientNet) against a scratch-trained baseline, weighting recall on defect classes so genuine faults are not missed, and applied Grad-CAM heatmaps to confirm the model attends to the actual defect region rather than background artefacts.

- Prototyped an inspection workflow in which drone- or handheld-captured images are scored automatically and ranked by severity, showing how CV-assisted screening can shorten inspection cycles and support risk-based integrity planning.

Dataset: NEU Surface Defect Database (metal surface defects: cracks, patches, pitted surfaces, scratches) with open corrosion-detection imagery, used as a proxy for pipeline surface condition, since no public, labelled inspection-image dataset of Nigerian pipeline assets currently exists.

Tools & Methods: Python, PyTorch / TensorFlow, OpenCV, convolutional neural networks, transfer learning, image augmentation, Grad-CAM explainability

3. AI-Based Fraud / Ghost Transaction Detection for Energy-Sector Finance

Objective: To address financial-integrity risk in energy-sector finance by detecting fraudulent or “ghost” transactions, drawn directly from hands-on exposure to NMDPRA’s Budget and Account System Control function.

- Profiled transactional and account data for duplicate, ghost, and anomalous entries, applying anomaly-detection and classification techniques that balance detection sensitivity against false-positive rate.
- Structured findings into a reviewable output for finance and audit stakeholders rather than a black-box score, outlining a framework to strengthen internal financial controls and transaction integrity.

Dataset: NEITI oil & gas payment/receipt reconciliation data for sector context; the PaySim synthetic transactions dataset (Kaggle) used as a proxy to develop and validate the methodology, as no public transaction-level energy-sector fraud dataset exists.

Tools & Methods: Python, statistical outlier detection, supervised classification

4. Energy Asset Digital Twin / Monitoring Platform

Objective: To prototype a digital-twin concept for near-real-time monitoring of energy assets (e.g., storage or processing equipment) in order to improve operational visibility.

- Modelled asset-status data (operational state, throughput, condition indicators) and combined it with dashboard visualization to represent asset health and status in a single view.
- Showed how a digital-twin layer could integrate with the anomaly-detection and forecasting work above, reducing information lag between field assets and decision-makers.

Dataset: Volve Field Open Dataset (Equinor) real-time drilling and production sensor logs, used to represent asset operational state and condition indicators.

Tools & Methods: Python, dashboard/BI tooling, data modelling

5. Statistical Research Study - B.Sc. Thesis

Objective: Design and execute a data-driven study analyzing the causes and prevention of examination malpractice using inferential statistics and survey data.

- Designed the data-collection instrument, gathered primary survey data, applied inferential statistical tests, and translated the results into clear, actionable recommendations.

Tools & Methods: Statistical analysis, survey design, inferential testing

EDUCATION

B.Sc. (Ed.) in Mathematics & Education

2020 – 2025

University of Uyo - Akwa Ibom, Nigeria

Second Class Honours (Upper Division). Thesis: a statistical study of the causes and situational prevention of examination malpractice in school mathematics.

Relevant Coursework: predictive modelling, statistical analysis & probability, data analytics, optimization, operations research, numerical analysis, computational mathematics.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

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|---|------------------------|
| • Professional Certificate in Data Science – VEXTOR HUB LIMITED | Feb. 2025 – May 2025 |
| • Professional Certificate in AI/ML Engineering – VEXTOR HUB LIMITED | July 2025 – Nov 2025 |
| • National Youth Service Corps (NYSC) Discharge Certificate | Jan. 2026 – Dec. 2026 |
| • Leadership & Management Certificate – More Leadership Academy, Uyo | Sept. 2023 – Dec. 2023 |
| • Certified Teacher – Teachers Registration Council of Nigeria – TRCN | Aug. 2025 |
| • Smfest Volunteering Certification | Oct. 2025 |

LEADERSHIP & ADDITIONAL EXPERIENCE

Selected roles demonstrating stakeholder engagement, reporting, and project coordination, transferable to regulatory, corporate, and multilateral energy-sector environments.

Head of Media & Technical Department

May 2025 – Apr 2026

Destiny Centre Unveiling Chapel

- Led the church's Media & Technical Department, overseeing live sound, live-streaming, and audio-visual delivery for all services, and coordinated the team and digital content production.

Program & Digital Media Support

Sep 2024 – Dec 2025

DipGold Human Development Foundation (DGHDF)

- Supported monitoring & evaluation for community-development programmes, capturing data on activities, reach, and outcomes, and assisted with field data collection and digital media content alongside community stakeholders.

President, Students' Union Government

Jan 2023 – Dec 2023

Akwa Ibom State University of Education (Affiliated to University of Uyo)

- Led the union's executive council, represented the student body to university management, and oversaw union budgets, translating student needs into measurable programmes.

Senator, National Students' Parliament

Jan 2023 – Dec 2023

National Association of Nigerian Students (NANS)

- Represented constituent students in the national parliament, participating in deliberations and votes on student-affairs policy, and contributed to committee work and oversight of national NANS programmes.

Convener & Lead Coordinator

2023 – Present

Synergy Conference

- Conceived and organised a multi-sector engagement conference end-to-end, from concept and speaker/partner sourcing through logistics and on-ground execution, and produced post-event reporting for stakeholders.

Member, Environmental Protection & Sanitation CDS

Jan 2026 – Dec 2026

National Youth Service Corps (NYSC)

- Planned and executed community environmental and sanitation projects under NYSC's Community Development Service, monitored outcomes, and developed outreach materials.

Member

2025 – Present

YALI Nigeria · American Spaces Nigeria · Smfest Africa

- Active participant in leadership-development, digital-literacy, and innovation programmes, building a network across youth-leadership and cross-cultural ecosystems in Nigeria and Africa.

LANGUAGES

English (Full Professional / Native-level) • Ibibio (Native) • French (Basic) • German (Basic)

INTERESTS & ACTIVITIES

- Coding and energy/automation personal projects; travelling, cross-cultural collaboration, and cultural-heritage reading.